

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

(5 Marks)

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

(5 Marks)

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.

(5 Marks)

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs:

(5 Marks)

- (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
- (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
- (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
- (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
- (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
- (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
- (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
- (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.

- ii) Suggest two effective techniques to reduce overfitting in this model.
- iii) Discuss how overfitting affects the model's generalization ability on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

- (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
 - (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
 - (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
 - (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
 - (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
 - (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
 - (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
 - (h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%
- i) Identify the epoch at which the model starts overfitting.
 - ii) Explain why validation accuracy decreases despite training accuracy increasing.
 - iii) Suggest two methods to improve validation performance.

①
sol:-

Given

Box	Red	blue	Green	total
A	6	4	2	12
B	5	7	3	15
C	8	2	5	15

Box B is twice as likely as A and C.

Therefore,

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{1}{2}, \quad P(C) = \frac{1}{4}$$

Probability of blue:

$$P(B/A) = \frac{4}{12} = \frac{1}{3}$$

$$P(B/B) = \frac{7}{15}$$

$$P(B/C) = \frac{2}{15}$$

By using Baye's theorem:

$$P(C/B) = \frac{P(C) P(B/C)}{P(A) P(B/A) + P(B) P(B/B) + P(C) P(B/C)}$$

$$= \frac{\frac{1}{4} \times \frac{2}{15}}{\left(\frac{1}{4} \times \frac{1}{3}\right) + \left(\frac{1}{2} \times \frac{7}{15}\right) + \left(\frac{1}{4} \times \frac{2}{15}\right)}$$

$$= \frac{\frac{2}{60}}{\frac{1}{12} + \frac{7}{30} + \frac{2}{60}} = \frac{\frac{2}{60}}{\frac{5+14+2}{60}}$$

$$= \frac{2/60}{\frac{21}{60}} \Rightarrow \frac{2}{60} \times \frac{60}{21}$$

$$P(C/B) = \frac{2}{21}$$

Given

Box	Red	blue	black	total
A	5	3	2	10
B	4	6	5	15
C	7	2	6	15

B is twice as likely as A and C.

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{1}{2}, \quad P(C) = \frac{1}{4}$$

Blue probabilities:-

$$P(B|A) = 3/10$$

$$P(B|B) = 6/15 = \frac{2}{5}$$

$$P(B|C) = 2/15$$

By using Bayes's Theorem

$$P(C/B) = \frac{\frac{1}{4} + \frac{2}{15}}{\left(\frac{1}{4} \times \frac{3}{10}\right) + \left(\frac{1}{2} \times \frac{2}{5}\right) + \left(\frac{1}{4} \times \frac{2}{15}\right)}$$

$$= \frac{\frac{2}{60}}{\frac{3}{40} + \frac{2}{10} + \frac{2}{60}}$$

$$= \frac{\frac{2}{60}}{\frac{9+24+4}{120}} = \frac{\frac{2}{60}}{\frac{37}{120}}$$

$$\frac{2}{60} \times \frac{128^2}{37}$$

$$= 4/37$$

$$P(C/B) = 4/37$$

3.
sol.

Given,

There are 1800 samples per class

There are 2 classes so

$$1800 + 1800 = 3600$$

20% is reserved for testing

$$3600 \times \frac{20}{100} = 720$$

So, total test samples = 720

Given,

$$T_p = 210, \quad T_N = 190, \quad F_p = 30, \quad F_N = ?$$

We know,

$$T_p + T_N + F_p + F_N = 720$$

$$210 + 190 + 30 + F_N = 720$$

$$430 + F_N = 720$$

$$F_N = 290$$

Step 2:- Calculate Accuracy

Formula:

$$\text{Accuracy} = \frac{T_p + T_N}{T_p + T_N + F_p + F_N} \times 100$$

$$= \frac{210 + 190}{720} \times 100$$

$$= \frac{400}{720} \times 100$$

$$\boxed{\text{Accuracy} = 55.56\%}$$

Step 3:- Calculate Precision

$$\text{Precision} = \frac{TP}{TP + FP} \times 100$$

$$= \frac{210}{210 + 30} \times 100$$

$$= \frac{210}{240} \times 100$$

$$\boxed{\text{Precision} = 87.5\%}$$

Step 4:- Calculate Sensitivity / Recall

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100$$

$$= \frac{210}{210 + 290} \times 100$$

$$= \frac{210}{500} \times 100$$

$$\boxed{\text{Sensitivity} = 42\%}$$

Step 5:- Calculate Specificity

$$\text{Specificity} = \frac{TN}{TN + TP} \times 100$$

$$= \frac{190}{190 + 30} \times 100$$

$$= \frac{190}{220} \times 100$$

$$\boxed{\text{Specificity} = 86.36\%}$$

4
Sol

Training and Validation loss

Epoch	Training loss	Validation loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

i) Epoch at which overfitting begins

Validation loss is minimum at Epoch 3.

From Epoch 4, training loss decreases but validation loss increases.

Ans:- Overfitting begins at Epoch 4.

(ii) Two methods to reduce overfitting

1. Dropout
2. Early stopping

Other methods:- regularization, data augmentation, reducing model complexity.

(iii) Effect on generalization.

Overfitting causes the model to memorize training general patterns.

$\therefore$ Therefore,

Training performance $\uparrow$

Unseen-data performance $\downarrow$

5. Training and Validation Accuracy

Epoch	training Accuracy	validation Accuracy
1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

(i) When does Overfitting start?

Validation accuracy reaches its highest value of 73% at epoch 5. After that, validation accuracy decreases while Overfitting starts from epoch 6.

(ii) Why does validation accuracy decrease?

The model starts memorizing the training data rather than learning general patterns.

Therefore:

- Training accuracy increases: 80% $\rightarrow$ 92%
- Validation accuracy decreases: 73% $\rightarrow$ 68%

(iii) Two methods to improve validation performance.

1. Early stopping - stop training validation performance accuracy (epoch 5).

2. Dropout / Regularization - prevents excessive dependence on particular features.